

[Geolocked]

Carbon dioxide removal prepurchase application 2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the [Frontier GitHub repository](#) after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Geolocked

Company or organization location (we welcome applicants from anywhere in the world)

Houston, TX

Name(s) of primary point(s) of contact for this application

LoongYi Tan

Brief company or organization description <20 words

Permanent carbon sequestration via deep well waste biomass injection

1. Public summary of proposed project¹ to Frontier

- a. **Description of the CDR approach:** Describe how the proposed technology removes CO₂ from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

Geolocked is a Houston-based company, sequestering bio-based carbon from biomass at scale. Our method and technology allows us to inject bio-based biomass waste in permitted and widely available underground Class I and Class II injection wells, minimizing regulatory permitting risk. This method

¹ We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

allows us to obtain an all in carbon disposal today at approximately \$300-350 / ton, after factoring labor, trucking, and injection costs. In the near future, Geolocked would also own and operate a network of distributed Class I and Class II wells, allowing us to obtain an all in disposal cost of <\$100 / ton.

The carbon products that will be injected will be organic waste from a number of different streams, including but not limited to biosolids from wastewater treatment plants, biomass waste from biorefineries such as pulp and paper plants, food processing plant waste, ethanol mills, SAF (Sustainable Aviation Fuel) plants, and renewable diesel / biodiesel plants.

As we are leveraging existing, permitted injection wells, this allows us to access over 32,000 permitted injection wells, which is significantly more than other injection methods such as salt dome injection. This dispersed method of injection allows us to be uniquely capable of meeting the target criteria of <\$100/ton of CO₂e and access different sources of waste biomass. The method also allows our method to be feedstock agnostic, and to solve for carbon removal methods that will have the largest impact at the lowest cost.

By starting with waste biomass, this will allow us to capture and sequester carbon at the lowest cost due to a number of factors:

- All biomass can trace its roots through photosynthesis, which is the most effective method of removing carbon from the atmosphere.
- Waste biomass is being diverted away from landfills and into injection wells which reduces methane generation from anaerobic digestion, whether intentional (via renewable gas generation) or not.
- Landfill diversion also has other ecological benefits, as diversion away from landfills allow for PFAS leachate production to be minimized.
- Landfill diversion has social & environmental justice benefits as well, since landfills are often located in low-income communities largely known as “sacrifice zones”.
- As these products are end-of-life, these wastes allow our feedstock providers to avoid a tipping fee. In some instances, the feedstock providers may pay Geolocked to take the waste away from their facility.

The use of distributed, permitted underground Class I and Class II injection wells allow for the following benefits:

- Minimizing transport / trucking costs and its environmental emissions.
- Leveraging an existing network of logistics and labor who are already familiar with the feedstock as well as the injection network.
- Utilizing off the shelf, capex-light processing methods that fit the biomass source to the injection site.
- For example, the Texarkana-Louisiana area has a high concentration of pulp and paper plants. By utilizing well known and off the shelf processing equipment such as grinders and milling equipment, it will allow us to inject these biomass waste streams into the formation without the use of experimental, capital and energy intensive methods such as pyrolysis or hydrothermal liquefaction.

Beyond biomass, this disposal-injection method can have other co-benefits beyond carbon removal. The use of injection wells has been widely regarded and accepted as one of the most permanent and safest methods of disposal. In fact, many landfills utilize this method to dispose of their landfill leachate (which contain high levels of forever chemicals such as PFAS & PFOS). This distributed injection method will also allow for the disposal of these forever chemicals commonly found in municipal wastewater sludge in a permanent and safe manner.

https://www.epa.gov/sites/default/files/2015-07/documents/study_uic-class1_study_risks_class1.pdf

<https://www.cdmsmith.com/en/client-solutions/insights/pfas-landfill-leachate>

<https://www.epa.gov/biosolids/and-polyfluoroalkyl-substances-pfas-sewage-sludge>

<https://www.nytimes.com/2025/04/21/climate/sludge-fertilizer-farmland-superfund.html>

- b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?² What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

Geolocked is a relatively early company, being incorporated in January 2025. We envision a world where landfills have no eyesore, where the footprint of a waste disposal process is measured in square feet, not acres. We strive to build, own, and operate a distributed network of permitted injection wells across the US near sources of waste biomass.

In the near future, we will achieve ~\$300/t by leveraging an existing network of waste biomass and an injection well in the Texas Panhandle area. In that area, there are a large network of feedlots (providing agricultural waste), ethanol mills, fat rendering facilities, and biodiesel & renewable diesel plants. Our first injection facility in the panhandle is within a 3 hour drive of these facilities.

Our pathway of achieving <\$100/t in 2026 involves: 1) acquiring a disposal well network to allow us to process more forms of waste biomass, 2) colocate future disposal wells in regions where there is projected to have 5000-10000 tons of waste per year generated, and 3) optimize the chemical treatment and operations of these disposal sites to operation cost. The injection of biomass in deep geologic reservoirs has been seen as the most permanent and economic method of carbon removal in the near future.

One of the challenges that we have identified with injection is the need for water as a carrier fluid. By using a disposal well network, we will have access to water that folks will pay us to inject (at approximately \$1-6/bbl), which solves the water problem, as well as acting as a natural hedge and subsidy from the fossil fuel industry. We believe that these additional revenue streams from disposing wastewater waste such as those from the fossil fuel industry will be important in demonstrating the commercial resilience of the Geolocked business in the future.

- c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Market risk: Our technology has the capability to scale rapidly, as we are using existing supply chains, injection assets, and waste biomass sources. While promising, there needs to be a greater number of buyers who will purchase high-quality carbon credits at prices of >\$100/MT of CO₂e substantially in the coming years.

We have the ability to mitigate against this risk to some extent because our network of injection assets will have the ability to generate waste management revenues through the disposal of biomass, and oil and gas wastes as an alternative to landfills.

Permitting risk: Our technology requires the use of permitted underground injection sites to store biomass and organic wastes. The timelines can vary widely, as different states have primacy (regulatory jurisdiction) for these permits. As a result, obtaining permits vary widely from months to

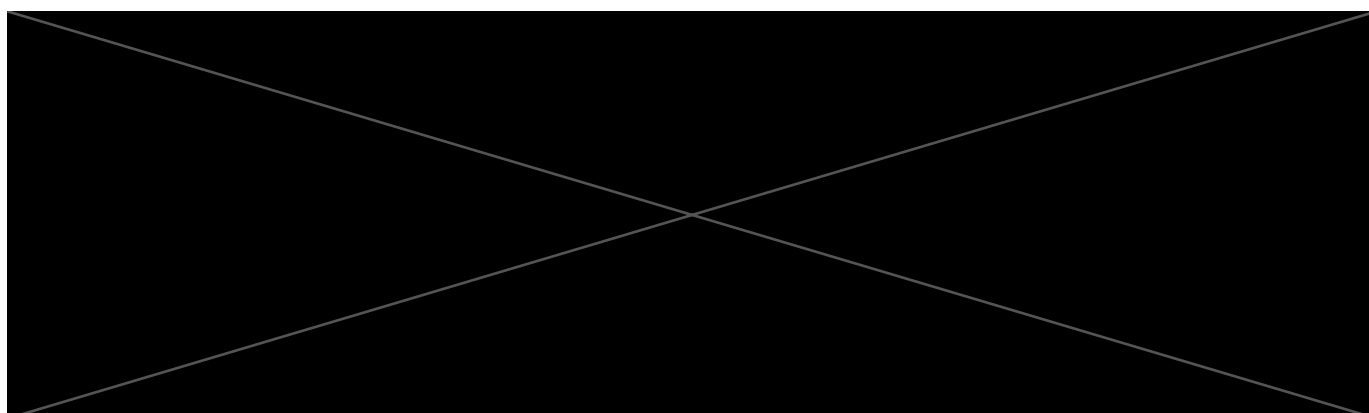
² We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at \$100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

years. This is a risk that all biomaterial injection CDR companies face but we plan to we plan to mitigate this risk by using more available class I and II injection wells which typically have a short permitting timeline of 2-3 months versus the 1-3 year timelines associated with salt cavern based wells that are favored by other biomaterial injection CDR companies.

Biomass availability: Our technology relies on waste biomass and organic waste sources to be available for us to inject underground to sequester carbon. However, there has been recent interest in waste biomass for other uses, such as for biogas and bio-natural gas production through anaerobic digestion facilities, which will lead to competition for locations near large waste biomass generators (>100k tons/yr). This is a risk that is inherent to all biomaterial injection CDR companies but potentially magnified for those who rely on large waste sources to be close by. We have mitigated this risk to some extent by developing a network of smaller distributed injection wells that will target smaller and sparser waste biomass sources that likely wouldn't be economical for other technologies to use and are projected to be plentiful based on the above analysis.

- d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) <i>(should be net volume after taking into account the uncertainty discount proposed in 5c)</i>	1536 tons CO ₂ e (2% uncertainty discount, 1566.85 net tons CO ₂ e) (equivalent to 1666.66 tons CO ₂ e of gross removals)
Delivery window <i>(at what point should Frontier consider your contract complete? Should match 2f)</i>	By December 31, 2025
Levelized cost (\$/ton CO ₂) <i>(This is the cost per ton for the project tonnage described above, and should match 6d)</i>	\$304
Levelized price (\$/ton CO ₂) ³ <i>(This is the price per ton of your offer to us for the tonnage described above)</i>	\$300



³ This does not need to exactly match the cost calculated for "This Project" in the TEA spreadsheet (e.g., it's expected to include a margin and reflect reductions from co-product revenue if applicable).